

HYPERTEXT MARKUP LANGUAGE

HTML Tables

HTML tables are used to display information in a structured grid made up of rows and columns. Tables are especially useful when data needs to be compared, organized, or presented in a format where relationships between values are easy to understand.

A table can contain text, numbers, images, links, or even other HTML elements inside its cells.

Why Use Tables?

Tables help to:

- Organize large amounts of information
- Display comparisons clearly
- Present schedules and timetables
- Show reports and statistics
- Structure tabular data logically
- Improve readability of related information

Real-Life Examples

Tables are commonly used for:

- Student mark sheets
- Product price lists
- Employee records
- Class timetables
- Financial reports
- Sports scoreboards

Table Basics

An HTML table is built using several tags that work together.

Main Table Elements

Tag	Purpose
<table>	Creates the table
<tr>	Creates a row
<th>	Creates a heading cell
<td>	Creates a data cell
<caption>	Adds a table title



Basic Table Structure

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>20</td>
  </tr>
</table>
```

Visual Representation:

Name	Age
John	20

Understanding the Structure

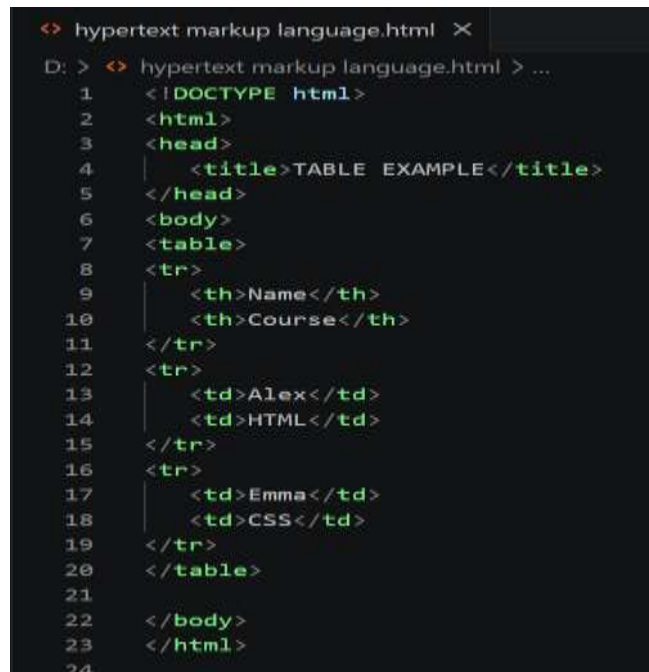
```
table
├── tr
│   ├── th
│   └── th
└── tr
    ├── td
    └── td
```

Where:

- Table contains rows.
- Rows contain cells.
- Cells contain content.

Adding Multiple Rows

```
<table>
<tr>
  <th>Name</th>
  <th>Course</th>
</tr>
<tr>
  <td>Alex</td>
  <td>HTML</td>
</tr>
<tr>
  <td>Emma</td>
  <td>CSS</td>
</tr>
</table>
```



```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>TABLE EXAMPLE</title>
5  </head>
6  <body>
7  <table>
8  <tr>
9  |   <th>Name</th>
10 |   <th>Course</th>
11 </tr>
12 <tr>
13 |   <td>Alex</td>
14 |   <td>HTML</td>
15 </tr>
16 <tr>
17 |   <td>Emma</td>
18 |   <td>CSS</td>
19 </tr>
20 </table>
21
22 </body>
23 </html>
24
```



Name	Course
Alex	HTML
Emma	CSS

Table Caption

A caption provides a title for the table.

```
<table>
<caption>Student Information</caption>
<tr>
  <th>Name</th>
  <th>Age</th>
</tr>
</table>
```

```

hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>TABLE EXAMPLE</title>
5  </head>
6  <body>
7  <table>
8  <caption>Student Information</caption>
9  <tr>
10 |   <th>Name</th>
11 |   <th>Age</th>
12 </tr>
13 </table>
14 </body>
15 </html>

```

TABLE EXAMPLE	X	+
 File D:/hypertext%20markup%20language.html		
Student Information Name Age		

Rowspan and Colspan

Sometimes data must span across multiple rows or columns.

HTML provides:

- rowspan
- colspan

for this purpose.

What is Rowspan?

rowspan allows one cell to extend vertically across multiple rows.

Example

```

<table border="1">
<tr>
  <th>Name</th>
  <th>Subject</th>
</tr>
<tr>
  <td rowspan="2">David</td>
  <td>HTML</td>
</tr>
<tr>
  <td>CSS</td>
</tr>
</table>

```



```

hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>TABLE EXAMPLE</title>
5 </head>
6 <body>
7 <table border="1">
8 <tr>
9 | <th>Name</th>
10 | <th>Subject</th>
11 </tr>
12 <tr>
13 | <td rowspan="2">David</td>
14 | <td>HTML</td>
15 </tr>
16 <tr>
17 | <td>CSS</td>
18 </tr>
19 </table>
20 </body>
21 </html>
22

```

TABLE EXAMPLE

File D:/hypertext%20markup%20language.html

Name	Subject
David	HTML
	CSS

The "David" cell occupies two rows.

What is Colspan?

colspan allows one cell to extend horizontally across multiple columns.

Example

```

<table border="1">
<tr>
  <th colspan="2">
    Student Details
  </th>
</tr>
<tr>
  <td>Name</td>
  <td>John</td>
</tr>
</table>

```

```

hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>TABLE EXAMPLE</title>
5 </head>
6 <body>
7 <table border="1">
8 <tr>
9 | <th colspan="2">
10 |   Student Details
11 | </th>
12 </tr>
13 <tr>
14 | <td>Name</td>
15 | <td>John</td>
16 </tr>
17 </table>
18 </body>
19 </html>
20

```

TABLE EXAMPLE

File D:/hypertext%20markup%20language.html

Student Details	
Name	John

Combining Rowspan and Colspan

Both attributes can be used together.

```
<table border="1">
<tr>
  <th rowspan="2">Name</th>
  <th colspan="2">Marks</th>
</tr>
<tr>
  <th>Math</th>
  <th>Science</th>
</tr>
<tr>
  <td>Alex</td>
  <td>85</td>
  <td>90</td>
</tr>
</table>
```

```
<> hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>TABLE EXAMPLE</title>
5  </head>
6  <body>
7  <table border="1">
8  <tr>
9  |   <th rowspan="2">Name</th>
10 |   <th colspan="2">Marks</th>
11 </tr>
12 <tr>
13 |   <th>Math</th>
14 |   <th>Science</th>
15 </tr>
16 <tr>
17 |   <td>Alex</td>
18 |   <td>85</td>
19 |   <td>90</td>
20 </tr>
21 </table>
22 </body>
23 </html>
```

Name	Marks	
	Math	Science
Alex	85	90



Table Sections

Large tables can be divided into sections.

`<thead>`

Contains table headings.

`<thead>`

`<tr>`

`<th>Name</th>`

`</tr>`

`</thead>`

`<tbody>`

Contains main table data.

`<tbody>`

`<tr>`

`<td>Alex</td>`

`</tr>`

`</tbody>`

`<tfoot>`

Contains summary information.

`<tfoot>`

`<tr>`

`<td>Total</td>`

`</tr>`

`</tfoot>`

Complete Example:

```
<> hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>TABLE EXAMPLE</title>
5  </head>
6  <body>
7    <table>
8      <thead>
9        <tr>
10         <th>Item</th>
11         <th>Price</th>
12       </tr>
13     </thead>
14     <tbody>
15       <tr>
16         <td>Book</td>
17         <td>$20</td>
18       </tr>
19     </tbody>
20     <tfoot>
21       <tr>
22         <td>Total</td>
23         <td>$20</td>
24       </tr>
25     </tfoot>
26   </table>
27 </body>
28 </html>
```

TABLE EXAMPLE X +

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Item Price
Book \$20
Total \$20

